CS 7650 NLP Class Jupyter Notebook ICE Support

aka How To Find ICE On The Moon

version: 2.0

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NOTE: Running on Google Colab Pro starts from $10 dollars per month and gives you the ability to reserve A100 GPU with at least 100 credits which equates to about 10hr of GPU time. It is the simplest way to do the work for the majority of students. Using Jupyter notebook on ICE cluster machine with a GPU can be more difficult to setup and download models from. We highly encourage everyone to use Colab Pro option first. In some cases, it may be very difficult to schedule ICE machine during peak times and system maintenance.

How to Setup Jupyter Notebook(s) on ICE machine with a GPU

1. connect to VPN

2. go to this website: https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard

3. upload your HW files to directory of your choice (by going to Files->scratch->cs7650 (new folder, create it) here, no limit on storage for a semester)

4. Open a terminal on ICE (the easiest is **Clusters → ICE Shell Access**.

5. Create / activate your environment and install packages:

In bash, type to install python 3.12 version.

module load anaconda3

conda create -n myenv python=3.12 -y

conda activate myenv

Go to your project directory (cd scratch/cs7650) where you saved your files and do this (the requirements.txt file came with your homework). There should not be any errors.

Note: the requirements.txt file setup was tested without libraries' versions included

pip install --user -r requirements.txt

conda install ipykernel -y

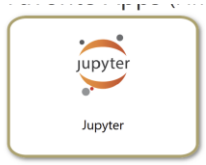
Register the kernel (still in the same terminal, with the conda environment activated):

python -m ipykernel install --user --name myenv --display-name "Python 3.12 (myenv) NP"

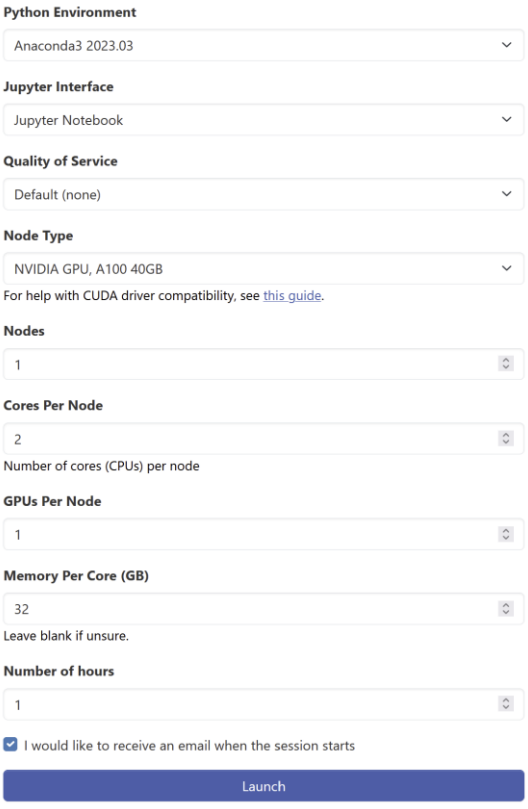
Reserving ICE Machine For Development

Reserve the Jupyter notebook machine by selecting Jupyter notebook icon on home page

(https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard)



And selecting (example config)



Note: For better resource availability you can choose: **Node Type** to be any first available GPU with CUDA v13, that way you will be able to register GPU sooner.

Note: Sometimes leaving **Memory Per Core (GB) blank** resolves BadConstraints on queue error.

Click Launch.

Click Connect to Jupyter notebook after it starts.

How To Download/ Upload Models/Files The Easiest Way From Jupyter Notebook that generated data.

1. connect to VPN
2. Go to https://ondemand-ice.pace.gatech.edu/pun/sys/dashboard
3. Open the Files menu at the top of the page.
4. Browse to your home or scratch directory where you save your files and models and download them.

Jupyter Notebook Adaptations To Run On ICE machine

- Always select Kernel --> Change Kernel **Python 3.12 (myenv) NP** The one that you created above.
- Update the **Project_ROOT** path: in notebook to: "." current folder where you saved your notebook(s). Local current directory that is which hopefully is ~/scratch/cs7650 (300GB disk space that is not backed up by ICE infrastructure).
- Disable google drive code in notebook because you will be saving into ICE machine folder (scratch folder, for example) for example.
- Current library setup requires a user to install libraries via notebook the following way:
  %pip install google-genai instead of **!pip install google-genai**
  This ensures that one uses python 3.12 that we just configured initially for ICE.
  Any references to **!pip need to be updated to %pip** for this ICE setup to work with current kernel.
- At the top of each Jupyter notebook file add the following lines for better debugging of your environment if you post on Ed:
  import sys
  print(sys.executable)
  print(sys.version)

  The result of these lines should show:

  ```
  /home/hice1/user_username/.conda/envs/myenv/bin/python
  ```

  ```
  if it does not, it means that you have not configured the environment correctly.it should show you that you have python 3.12 or better.
  ```

  ```
  Also add this line to your notebook.
  ```

  !cat ~/.local/share/jupyter/kernels/myenv/kernel.json

```
This will tell you if you have Jupyter kernel created correctly too.
```

- **Common Errors:** `OSError: [Errno 122] Disk quota exceeded:`
  `Solution: delete ~/.cache/ contents`

Also, for the first trainer model. We found that some solutions can converge as fast as 2 epochs, so you are welcome to update NUM_TRAIN_EPOCHS = 2 for the sack of debugging the training routine and put it back to default when submitting the notebook.

Starting Notebook

Now go to Open OnDemand → Interactive Apps → Jupyter, launch a session as usual, and select Kernel --> Change Kernel to **Python 3.12 (myenv) NP.** Now you can run it by clicking on “Run” mini button. Congrats you made it!